PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Can Incentives Improve Antipsychotic Adherence in Major Mental iIllness? A mixed-methods systematic review
AUTHORS	Hodson, Nathan; Majid, Madiha; Vlaev, Ivo; Singh, Swaran

VERSION 1 – REVIEW

REVIEWER	Zipursky, Robert CAMH, Schizophrenia Speaking fees and research support from Janssen Inc.
REVIEW RETURNED	17-Dec-2021

GENERAL COMMENTS	This paper addressed important issues related to the use of incentives to improve antipsychotic adherence. The paper reviews the evidence that incentives impact on antipsychotic adherence as well as clinical outcomes, and the ethical and practical issues that arise in providing incentives for antipsychotic adherence. The Background would benefit from substantial revisions so that the issues of incentives for the antipsychotic treatment of schizophrenia are cast in the appropriate perspective. The Background section begins with, "Some people prescribed antipsychotic do not take them." This is likely to be the case for all medications that are prescribed so the statement is not meaningful. It is important that the issue of adherence to antipsychotics in people with schizophrenia be appreciated in the context of the broader issues of adherence in Medicine. Antipsychotics are prescribed for many indications and it is likely that the minority of prescriptions are for people with schizophrenia given that medications like quetiapine are widely prescribed by general practitioners for insomnia. It would be valuable to be clear, ideally in the title itself, that this is a paper about incentives for the antipsychotic treatment of schizophrenia. The role of long-acting antipsychotics in the management of bipolar disorder is less well established at this time. Adherence to antipsychotics is described in the first paragraph as ranging from 60-81%. The reference is to a paper on adherence to antipsychotics in schizophrenia and bipolar disorder. Estimates of adherence rates in chronic illness across all area of medicine are estimated to approximately 50% (Haynes et al. 2020) which is consistent with estimates of adherence rates in people with schizophrenia. Problems with adherence are the single most important determinant of risk for relapse in schizophrenia which is associated with not only enormous health care costs but enormous distress and suffering for patients and their families. In addition,

	current research suggests that relapses may increase the risk of developing treatment resistance and poor outcomes. Preventing relapses is a key priority in the treatment of schizophrenia. As described in the paper, problems with adherence do not necessarily reflect opposition or ambivalence about medication but rather challenges in maintaining high and consistent levels of administration. Long-acting antipsychotics may address this in part and the addition of incentives may provide further improvements. In the second paragraph, it is stated that, "previous systematic reviews have found that depot treatment did not increase adherence compared to oral treatment and there is no difference in relapse rates in RCTs comparing long-acting injectable to oral preparations." This has been a very complex area of research for which RCTs are probably not the optimal methodology to assess effectiveness. The meta-analysis referenced by Kishimoto (2014) has been updated in 2021 and reported significant effects in favour of long-acting medication including in RCTs. Other methodologies including analyses of data from national cohort studies in Finland favour long-acting antipsychotics. The literature review should be updated in the revised manuscript. "DIALOG" and "MANSA" scores are not defined or explained in the text. This would be valuable to include in order to interpret the results of the Phase One analysis.

REVIEWER	Taub, Sharon Geha Mental Health Centre
REVIEW RETURNED	23-Jan-2022

GENERAL COMMENTS	I find the topic clinically- important. More background is needed regarding non- adherence concerns and the ethical issues (in the background section). Better clarification of the different diagnosis is needed in the text (appears mostly in the tables). The issue is relevant and the overview of the the behavioral theory is interesting.

VERSION 1 – AUTHOR RESPONSE

Comments	**Responses from authors**
-In the Abstract, please include the study search date(s).	Done
-Please revise the 'Strengths and limitations of this study' section of your manuscript (after the abstract). This section should contain up to five short bullet points, no longer than one sentence each, that relate specifically to the methods. The novelty, aims, results or expected impact of the study should not be summarised here.	Thank you I have redrafted the strengths and limitations section following your guidance.
-Please include, as a supplementary file, the precise, full search strategy (or strategies) for all databases, registers and websites, including any filters and limits used.	The full search strategy is now included as a supplementary file at point 3
Formatting Amendments (where applicable): - Please ensure to upload the supplementary file in PDF format.	Done.

Reviewer: 1 Dr. Robert Zipursky, CAMH Comments to the Author: This paper addressed important issues related to the use of incentives to improve antipsychotic adherence. The paper reviews the evidence that incentives impact on antipsychotic adherence as well as clinical outcomes, and the ethical and practical issues that arise in providing incentives for antipsychotic adherence.	Thank you we are pleased that the reviewer found this review interesting.
The Background would benefit from substantial revisions so that the issues of incentives for the antipsychotic treatment of schizophrenia are cast in the appropriate perspective.	Thank you we have followed the reviewers advice and carefully revised this section.
The Background section begins with, "Some people prescribed antipsychotic do not take them." This is likely to be the case for all medications that are prescribed so the statement is not meaningful. It is important that the issue of adherence to antipsychotics in people with schizophrenia be appreciated in the context of the broader issues of adherence in Medicine. Antipsychotics are prescribed for many indications and it is likely that the minority of prescriptions are for people with schizophrenia given that medications like quetiapine are widely prescribed by general practitioners for insomnia. It would be valuable to be clear, ideally in the title itself, that this is a paper about incentives for the antipsychotic treatment of schizophrenia. The role of long-acting antipsychotics in the management of bipolar disorder is less well established at this time.	Thank you we have changed the title and opening paragraph to highlight that we are discussing antipsychotics in the context of major mental illness.
Adherence to antipsychotics is described in the first paragraph as ranging from 60-81%. The reference is to a paper on adherence to antipsychotics in schizophrenia and bipolar disorder. Estimates of adherence rates in chronic illness across all area of medicine are estimated to approximately 50% (Haynes et al. 2020) which is consistent with estimates of adherence rates in people with schizophrenia.	Thank you for drawing this to our attention. We have incorporated this interesting comment into the manuscript.
Problems with adherence are the single most important determinant of risk for relapse in schizophrenia which is associated with not only enormous health care costs but enormous distress and suffering for patients and their families. In addition, current research suggests that relapses may increase the risk of developing treatment resistance and poor outcomes. Preventing relapses is a key priority in the treatment of schizophrenia. As described in the paper, problems with adherence do not necessarily reflect opposition or ambivalence about medication but rather challenges in maintaining high and consistent levels of administration. Long-acting antipsychotics may address this in part and the addition of incentives may provide further improvements.	Thank you for these comments which we have incorporated into the text and which have made this a stronger paper.
In the second paragraph, it is stated that, "previous systematic reviews have found that depot treatment did not increase adherence compared to	Thank you for these helpful comments. We have rewritten this section making this much more comprehensive and

oral treatment and there is no difference in relapse rates in RCTs comparing long-acting injectable to oral preparations." This has been a very complex area of research for which RCTs are probably not the optimal methodology to assess effectiveness. The meta-analysis referenced by Kishimoto (2014) has been updated in 2021 and reported significant effects in favour of long-acting medication including in RCTs. Other methodologies including analyses of data from national cohort studies in Finland favour long-acting antipsychotics. The literature review should be updated in the revised manuscript.	including the 2021 Kishimoto paper and the Finnish papers on the topic.
"DIALOG" and "MANSA" scores are not defined or explained in the text. This would be valuable to include in order to interpret the results of the Phase One analysis.	Thank you for pointing this out. MANSA is an acronym which we have expanded in the text. DIALOG is not an acronym but we have defined it. We have also added references to both.
Reviewer: 2 Dr. Sharon Taub, Geha Mental Health Centre Comments to the Author: I find the topic clinically- important. More background is needed regarding non- adherence concerns and the ethical issues (in the background section).	Thank you for this positive review. We have rewritten much of the background to expand the context of non-adherence. We have included comments about the ethical issues in the background without presenting the results.
Better clarification of the different diagnosis is needed in the text (appears mostly in the tables).	Thanks for this point – I have now specified the diagnoses in the text as well as in the tables.
The issue is relevant and the overview of the the behavioral theory is interesting.	Thank you I'm pleased the reviewer found this paper interesting and relevant.